A browser-based seismic observatory for regional earthquake catalogue comparison in North America

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Abstract

Earthquake catalogues differ in coverage, measurements and revision histories, complicating regional comparison and reuse. We present a browser-based seismic observatory connecting a computational system, documented inputs and reproducible descriptive results. Four frozen USGS, NRCan and SSN catalogues supply 287,936 source records. Explicit adapters preserve reported magnitude types, missing values and provenance without reconciling physical events. Verified static delivery supports maps, administrative selection, statistics, historical rankings, spatial-pair queries and exports without Docker or a database server. A 659-record USGS example returns mean magnitude 5.311; matched Canadian and Mexican land selections contain 15 and 53 records. A separate 36-event ComCat pilot identifies magnitude changes in 13 of 31 comparable retained endpoint pairs and four crossings of 5.5. Independent SQL agrees with 4,720 conditional retained projections. Executable evidence contracts separate captured observations from archive-dependent calculations and unsupported historical reconstructions. Results remain descriptive: completeness, source accuracy, physical independence and usability are not established, and no earthquake forecast or seismic-risk estimate is produced.

Keywords: earthquake catalogues; geovisualization; provenance; retained revisions; reproducibility; evidence contracts

1. Introduction

Regional earthquake analysis must address heterogeneous measurements, duplicates and magnitude compatibility (Weatherill et al., 2016); Mexican hazard studies additionally require completeness assessment and declustering (Sawires et al., 2019). Displaying several catalogues together does not perform those operations.

Our problem is how to consult heterogeneous records, compare regional selections and derive reusable information while retaining their interpretation limits. The observatory connects three contributions: browser computation, documented seismic inputs and traceable results. It inspects frozen observations rather than providing live monitoring.

ComCat supplies preferred summaries and versioned products (U.S. Geological Survey, n.d.a-c), whose evolution matters in practice (Thompson et al., 2020). Libcomcat, Timeline and INGV already provide history capabilities (U.S. Geological Survey, n.d.d,e; Pintore, 2026). Following resolution-aware query research (Sismanis et al., 2009; Gardezi and Bertossi, 2011), our evidence contracts bound calculations when archive completeness and historical preferences are unknown. Separately documented inputs extend Villa Vargas et al. (2026); Figure 1 connects sources, computation and outputs.

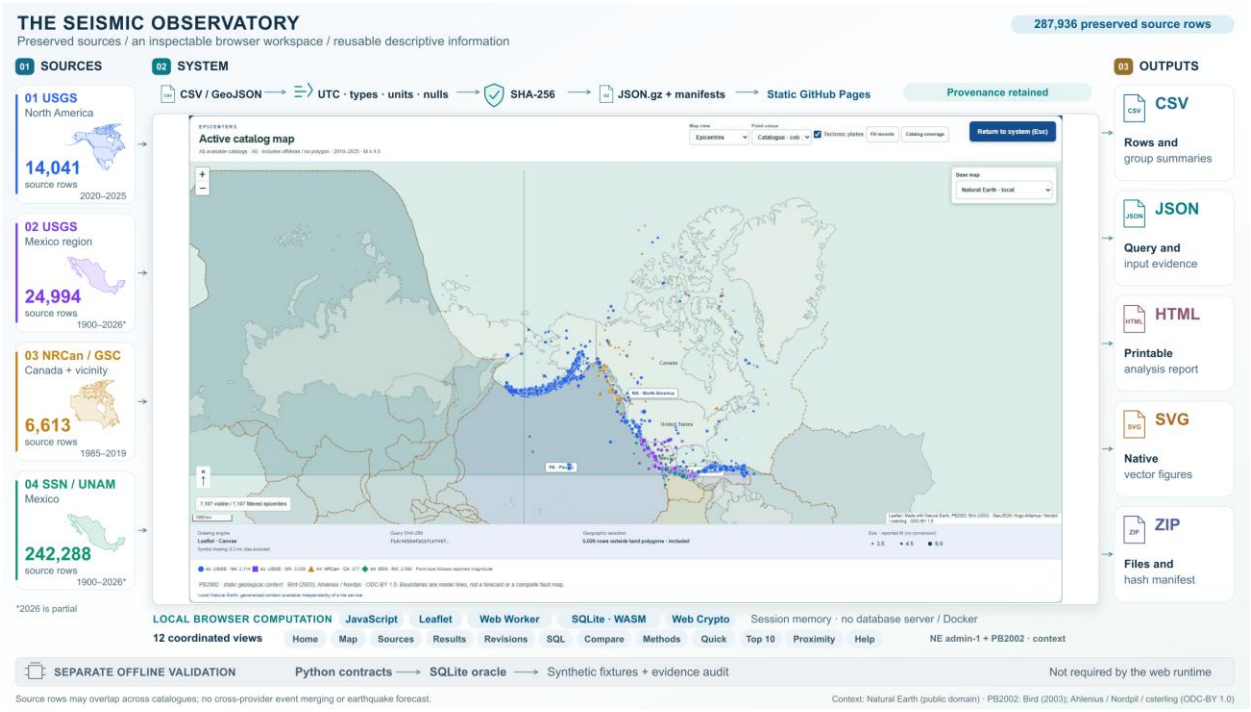


Figure 1. Implemented architecture connects four source packages, explicit adapters and verified static delivery to browser maps, analysis and reusable exports. The central map shows 7,107 selected rows (2010-2025, $M \geq 4.5$, all catalogues); inputs total 287,936. Optional SQLite/WebAssembly runs in a Worker; country/subdivision indexes and PB2002 supply separate context. The offline Python/SQLite audit is distinct from the browser runtime.

2. Computational system and analytical methods

2.1. Static architecture and reproducible delivery

Python adapters package preserved CSV inputs as JSON with SHA-256 manifests. GitHub Pages serves HTML, CSS, JavaScript, Leaflet 1.9.4 and bundled dependencies. Verified packages occupy browser session memory; no Docker, database server or persistent browser database is required. Home initially displays all 287,936 source rows (Figure 2). English is the default, with Spanish, themes and light/dark modes available.

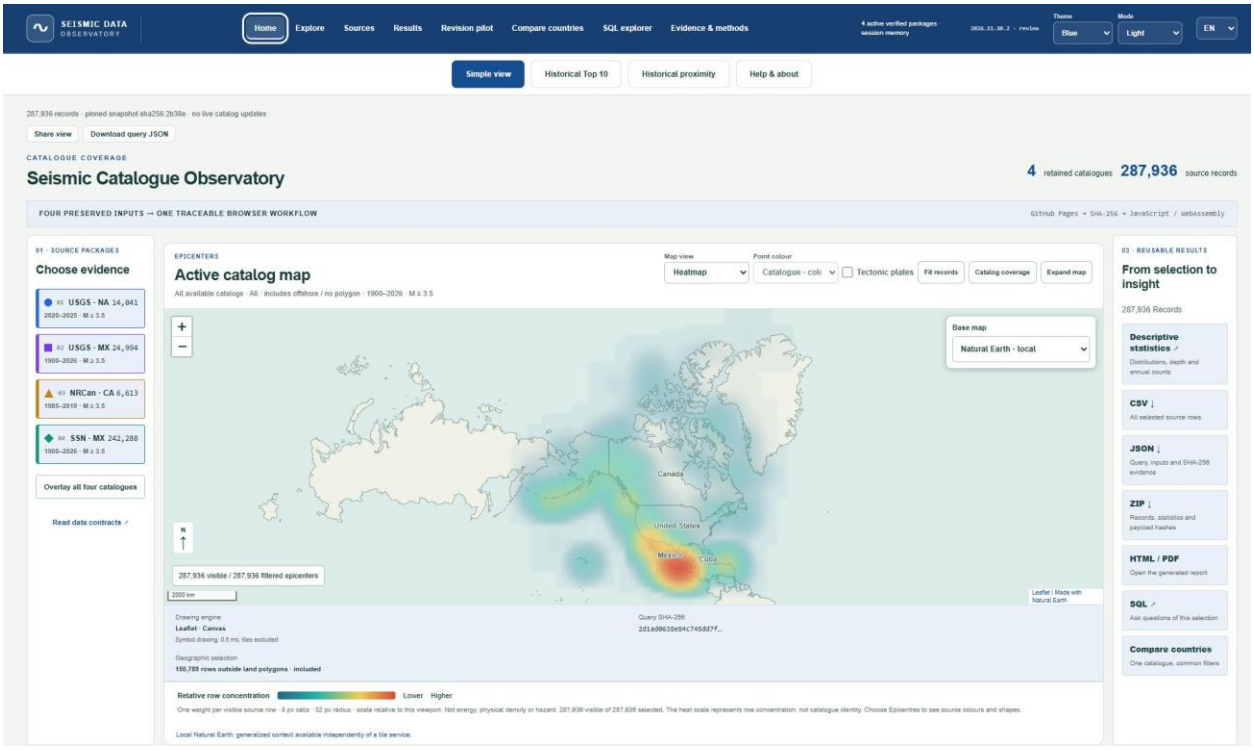


Figure 2. Home initially selects all 287,936 rows across four catalogues. Point and optional heat views retain the same query and exports. Heat displays relative screen-space row concentration, without magnitude weighting or hazard estimation. English, appearance controls and verification status remain visible.

Gzip companions bind compressed and original lengths/hashes, checked before decompression and JSON parsing. Bounded retries, progress and cooperative processing support loading. Unsupported gzip or HTTP 404/415 permits verified JSON fallback; integrity failures stop loading. Session caching avoids repeat work, without establishing speed or availability guarantees.

2.2. Normalization and geographic selection

Adapters align identifiers, explicitly labelled UTC origin times, coordinates, reported magnitudes and documented depths (Figure 3). They preserve case, source-empty strings, raw fields and unknown values; absent common fields become null. No magnitude conversion to Mw occurs. Canadian depth lacks documented units and stays outside kilometre statistics. SSN retains its UTC/local text. ComCat identifiers remain original; SSN/NRCan use content-hash row keys with occurrence suffixes. Atlas foregrounds source selection and the regional map; Time and Coverage aligns preserved origin-time intervals with map context. Both Sources views share package profiles and the same selection.

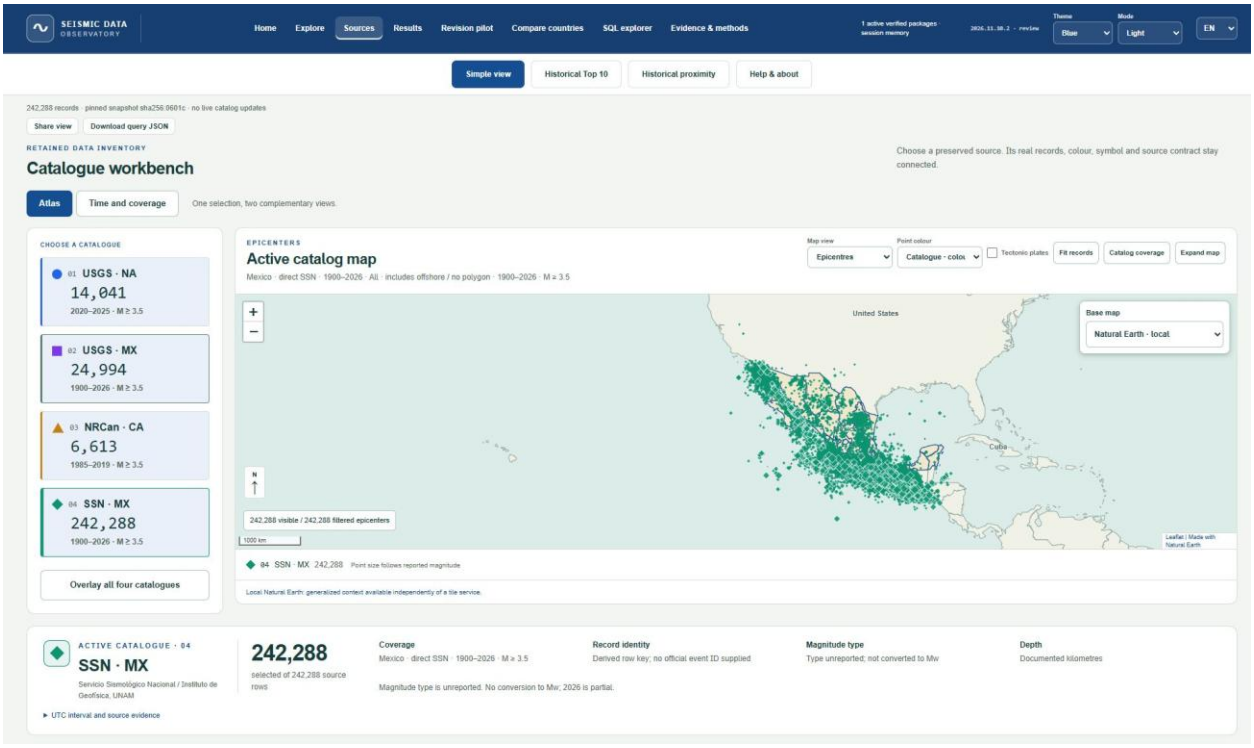


Figure 3. Sources Atlas displays the SSN package (242,288 rows), a regional map and a four-source rail. The complementary Time and Coverage view is accessible through its button. Counts, fields, verification and manifests retain each source's interpretation limits. Four packages from three distributors are not independent observing agencies.

Composition retains registry order and catalogue-qualified keys. It flags 2,163 shared ComCat IDs without merging rows or matching agencies' events by proximity. Thus 287,936 rows are not asserted independent earthquakes. Natural Earth supplies 47 generalized 1:50-million land classes and 743 pinned 1:10-million administrative units (Natural Earth, n.d.a,b). Mexico has 32 named units plus an unnamed minor-island unit. Derived indexes bind

geometry, event hashes and row order. Country clicks require confirmation before subdivision refinement. Offshore, boundary, invalid and unavailable classes remain explicit; containment establishes neither jurisdiction nor nationality. Mexican subdivision context can display all 33 units without filtering offshore rows (Figure 4).

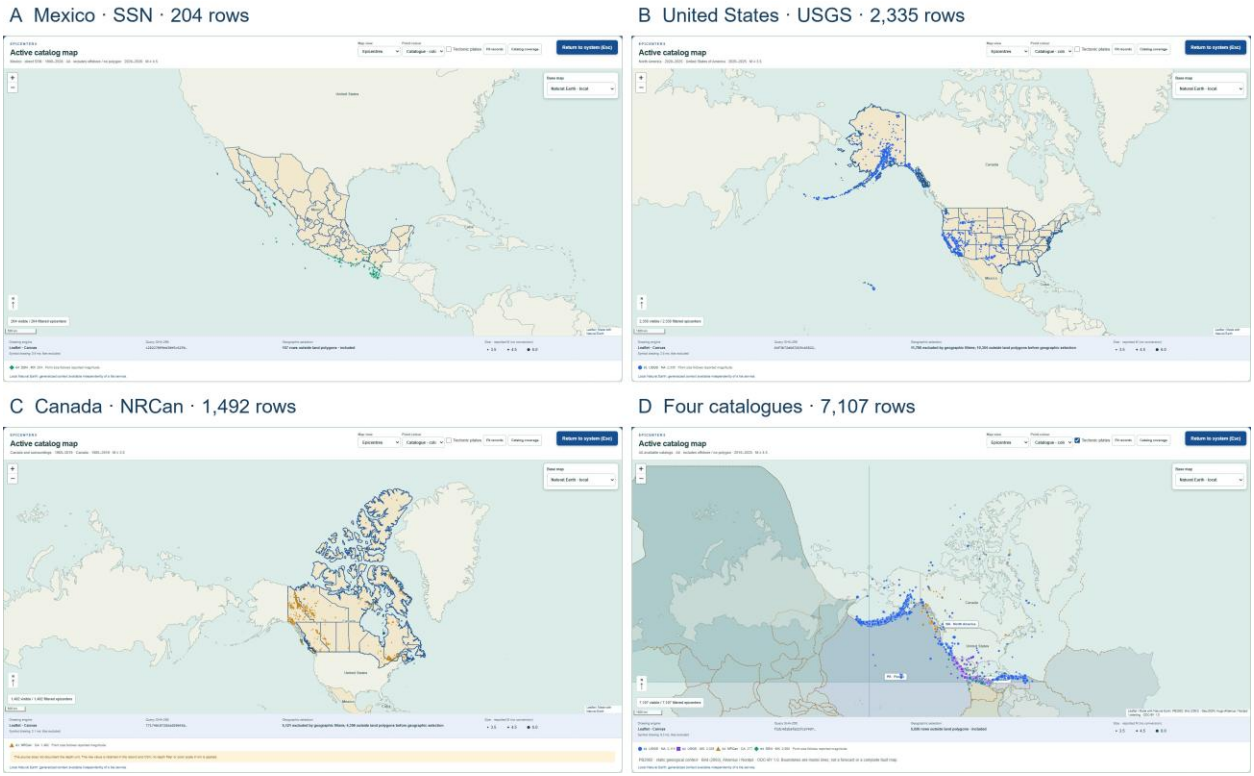


Figure 4. Four distinct map selections. (A) SSN, partial 2026, M ≥ 4.5, all classes: 204 rows, with Mexican subdivision context but no land filter. (B) USGS North America, 2020-2025, M ≥ 3.5, USA land: 2,335. (C) NRCan, 1985-2019, M ≥ 3.5, Canada land: 1,492. (D) All catalogues, 2010-2025, M ≥ 4.5: 7,107. Counts are source rows; map extent does not restrict exports.

Filters cover UTC years, reported magnitude/type, agency, documented depth, missing horizontal error and ID/place text. Coordinated views follow established practice (Wu et al., 2020). Local Natural Earth, OpenStreetMap, humanitarian OSM, USGS Topo and static August-2004 NASA Blue Marble provide backgrounds; attributed remote tiles load on demand. Map extent does not restrict exports. Antimeridian shifts affect display only. A query digest binds inputs, filters, geography and ordered keys, without supplying a digital signature.

Heat mode aggregates visible centres into 8-pixel CSS cells, one weight per row, with a Gaussian radius of 32 pixels and sigma 12 pixels. Log1p colours use the viewport maximum; edge kernels are clipped. Zoom and pan change display concentration, not physical density, energy or hazard. Points remain available and selections unchanged.

2.3. Descriptive tools and historical queries

Statistics use one catalogue and exact reported magnitude type; unknown types form a separate numerical group without asserted homogeneity. Means, medians, ranges and sample standard deviations retain their valid counts. Depth requires documented kilometres. Pearson correlation requires a known type, three finite pairs and nonconstant variables; unavailable values remain null. Annual zero bins mean no selected rows, not physical absence.

Country comparison applies common UTC years, threshold and type within USGS North America 2020-2025. Histograms share lower-inclusive/upper-exclusive 0.25-M bins; annual points retain discrete counts. No area/population normalization or equivalence of SSN/NRCan coverage is assumed.

Historical Top 10 defaults to all four catalogues with simple country/state filters and source/type on every row. It orders heterogeneous reported numbers, not a homogeneous magnitude scale or strongest-ever population. Exact ComCat IDs shared by the USGS packages produce one entry retaining both provenances and alternatives. Before filtering, the later updated timestamp selects the representative only if both timestamps are valid UTC; otherwise/ties prefer North America then Mexico, never the larger magnitude. SSN/NRCan rows are not physically matched. Single-catalogue/type/period selection remains advanced. A bounded selection retains only ten ranked candidates. Descending magnitude, ascending UTC time and key order at most ten entries, omitting further ties. Initial catalogue footprint retains offshore rows.

Historical proximity enumerates unordered pairs of distinct keys within one catalogue/type/UTC interval, default $M \geq 7$. Haversine uses Earth radius 6,371.0088 km and radii 25/50/100/200 km (default 100). Optional 100-km context admits unassigned/offshore rows near generalized land boundaries using minimum spherical point-to-minor-arc distance; assigned neighbours are not annexed. Both predicates allow 0.000000001 km numerical tolerance before rounding, not geographic uncertainty. Direct links imply neither transitive groups, repeating ruptures, recurrence intervals nor risk.

Optional PB2002 supplies static plate/boundary context (Bird, 2003; tectonicplates contributors, n.d.), without inferring mechanisms or active faults. Guided sql.js 1.13.0 SQLite/WebAssembly queries run in an isolated Worker (sql.js contributors, n.d.). Temporary relations preserve qualified keys and null identifiers/depths; query-only execution permits one SELECT/WITH statement, cancellation, timeout and 5,000 returned rows.

Simple view routes maps, statistics and reports through the same filter/export engine. Searchable Help explains sources, version, contact and licensing. CSV, JSON, HTML and SVG retain scope; ZIP includes payload

lengths/hashes. Printable HTML can save PDF. Query schema seismic-local-query/5 preserves administrative/geographic evidence; older supported URLs remain readable. These functions have computational checks, without an independent usability study.

2.4. Contracts for retained revisions

The separate offline audit identifies a lineage by (source, type, code) and a version by updateTime. Exact four-part repeats are deduplicated; conflicting contents fail. Distributor, origin author and magnitude author remain distinct. Primary endpoints require source us, type origin and code equal to the captured canonical event ID; public origins exclude internal-origin and phase-data products.

Table 1 separates captured-byte results, conditional retained projections and stronger historical claims. Projection first selects each lineage's latest retained producer time at or before a cutoff. A selected DELETE is reported separately without restoring its predecessor; attributes are then filtered. Reversing this order can retain obsolete above-threshold versions. Nulls fail value-dependent predicates; types retain case and magnitudes are not converted.

Table 1. Contract	Interpretation of the output	Principal boundary
`snapshot_observed`	Query result on supplied current summary bytes at capture	Not a historical replay or a complete physical population
`retained_projection_conditional`	Latest retained version per lineage at or before a producer-time cutoff	Neither public availability nor historical preference is established
`historical_catalog_not_identifiable`	Stronger historical catalog claim is declined	Missing historical evidence is not manufactured

Users acknowledge retained-only evidence and differences between producer/publication clocks. Unknown membership, release times, preferences and associations prevent reconstructing complete past public states. A refusal does not assert that a specific missing version existed.

Endpoints are earliest/latest retained non-deletion products, not first release/reference truth. Magnitude subtraction requires unchanged literal type. Inclusive thresholds are 5.5, 6.0, 6.5, 7.0, 7.5 and 8.0. Haversine displacement measures epicentral separation, not error improvement. Adjacent products are compared across 27 declared parameter, agency and uncertainty fields; unchanged selected fields do not prove identical complete products.

2.5. Independent evaluation

SQLite independently normalizes original products and ranks versions before filtering. Decimal parsing and integer UTC times provide a second implementation path. Ten query specifications test thresholds, literal mww/mb/mB, regional bounds, half-open 2017/2022 intervals and earthquake type, immediately before and at each retained update. Hashes and input lengths are checked.

Three synthetic paired histories share current evidence but have different past public release, preference or withdrawn membership. Eight further deterministic histories exercise missing revisions, deletion and clock/filter-order substitutions; a regional filter and three magnitude variants produce 24 cases. Independent public-action replay supplies known synthetic answers. These constructed controls test implementation and logical non-identifiability, not operational loss prevalence. Comparator documentation is inspected without performance or usability benchmarking; Timeline's maintained destination was inaccessible.

3. Seismic inputs and study design

Four catalogue packages from three distributors demonstrate exploration separately from revision validation (Table 2). Their coverage is not a sequence of historical public states.

Table 2. Catalogue and distributor	Origin-time coverage	Records	Interpretation limits
Mexico-region diagnostic / USGS ComCat	1900-2026 (partial)	24,994	Earthquake records; regional rectangle; two quarry blasts excluded; not an independent SSN catalogue
North American regional windows / USGS ComCat	2020-2025	14,041	Earthquake records from twelve pinned CSVs; regional windows, not national assignments or a revision archive
Canada and surroundings / NRCan/GSC	1985-2019	6,613	Reported $M \geq 3.5$; five repeats retained; original event IDs, event types and uncertainty/revision fields absent
Mexico and surroundings / direct SSN	1900-2026 (partial)	242,288	Reported $M \geq 3.5$ selected from 402,082 source rows; magnitude types, original IDs and revision fields absent; coverage is not completeness

Both Mexican packages have partial 2026 coverage. USGS North America combines latitude 14-84°, longitude -180 to -50°, with an Aleutian window at 50-73°, 170-180°; 770 positive-longitude records retain source coordinates.

The Canadian download of 6 November 2026 selects 6,613 M ≥ 3.5 rows from 101,365, retaining five exact repeats. Original identifiers, event types and undocumented depth units remain unknown. Open Government Licence-Canada applies to this resource, not automatically the NEDB bulletin (Natural Resources Canada, n.d.; Government of Canada, n.d.).

SSN was downloaded on 6 November 2026 for 1900-2026-11-06, all magnitudes/depths/states (Servicio Sismológico Nacional, 2026a). Its ten columns retain UTC/local date/time, magnitude, coordinates, kilometre depth, locality and status. Of 402,082 rows, 384,433 have numeric magnitudes and 17,649 have no calculable values; 242,288 satisfy M ≥ 3.5. Original IDs, magnitude/event types, uncertainties and revisions are absent; verificado/revisado do not supply version histories.

The source generation label nominally corresponds to 19:57:07 UTC, but the latest reported origin is 20:31:07 UTC. Eight UTC/local pairs have atypical offsets; neither column is corrected or the label treated as an atomic capture. Annual summaries preserve gaps and tied maxima. SSN terms and the original non-commercial reproduction notice retain source credit, meaning and third-party rights (Servicio Sismológico Nacional, 2026b). A documented 1988 reporting change and M_d 4.3 completeness for 1988-1998 (Zúñiga et al., 2000) do not establish completeness for this download.

The pinned upstream snapshot 8fe055ffe635875f7290906d45f70562a7896621 contains 24,996 unique IDs in 46 CSVs: 24,994 earthquakes and two excluded quarry blasts. UNM/us_unm origin-or-magnitude attribution selects 9,858 records. Contributor directories identify UNM with SSN/UNAM; USGS remains distributor (European-Mediterranean Seismological Centre, n.d.).

A separate ComCat pilot selects current earthquake records in 2017/2022 with preferred M ≥ 5.5, latitude 14-33° and longitude -118 to -86°. Retrieval on 5 November 2026 spans 19:53:15.316-19:53:54.609 UTC. Counts, truncation and duplicate IDs are checked; format=geojson/includesuperseded=true detail responses retain original bytes, URLs, retrieval intervals and hashes. Current selection omits previously qualifying events subsequently outside these criteria, so the cohort cannot estimate all historical withdrawals/crossings.

4. Results from reproducible selections

4.1. Regional information and reuse

Source profiles expose package coverage, fields, verification and downloads. Distinct map selections return 204 partial-2026 SSN rows at M ≥ 4.5, 2,335 USGS USA-land rows, 1,492 NRCan Canada-land rows, and 7,107 combined rows under their stated filters (Figure 4). Distributor choice and geographic containment remain separate. Figure 5 shows a guided SSN Guerrero land selection, UTC 2020-2025, M ≥ 3.5. Its 7,930 rows yield mean reported M 3.691, median 3.6 and mean documented depth 31.17 km. Summary measures and six annual bins share that selection; absent magnitude types do not establish a homogeneous scale.

A separate typed benchmark selects 659 USGS North America rows, UTC 2020-2025, M ≥ 4.5, literal mww and all geographic classes. Mean magnitude is 5.311168, median 5.2, range 4.5-8.2 and sample standard deviation 0.523513. All depths are documented: mean 32.832339 km, median 15.14 km. Pearson r is -0.003260, without establishing independence or causality. Python independently agrees with JavaScript and ordered identifiers within absolute tolerance 10^{-12}. Exports retain selection, criteria and hashes.



Figure 5. Guided descriptive results for 7,930 SSN Guerrero land records, UTC 2020-2025, M ≥ 3.5. Summary measures and six annual bins share the same selection. Reported magnitude types are absent; numerical grouping establishes neither scale homogeneity, completeness nor physical independence. No occurrence-rate or risk estimate is presented.

Administrative selection changes the query; plate display does not. Zoomed geographic context makes those separate controls visible (Figure 6). Neither supplies location uncertainty or mechanisms.

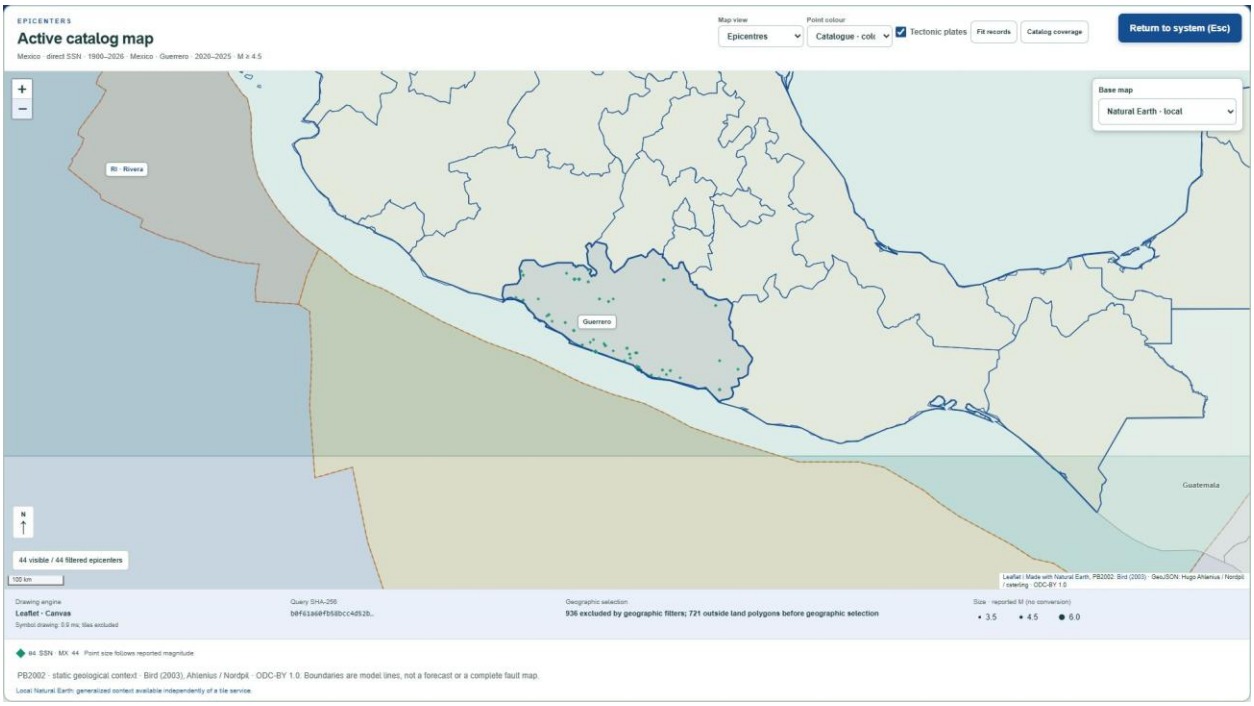


Figure 6. A regional map enlarges Guerrero/Oaxaca administrative limits, SSN symbols, scale and static PB2002 context. Confirmed country/subdivision filters alter the query; plate display does not reclassify records or supply mechanisms. Natural Earth classes are generalized land containment, with unassigned/offshore rows retained separately.

Under the same USGS period, threshold and mww type, Canadian and Mexican land classes contain 15 and 53 rows (Figure 7). Shared bins and annual counts describe selected records; unequal areas, offshore exclusions and unassessed completeness preclude relative-hazard or occurrence-rate comparisons.

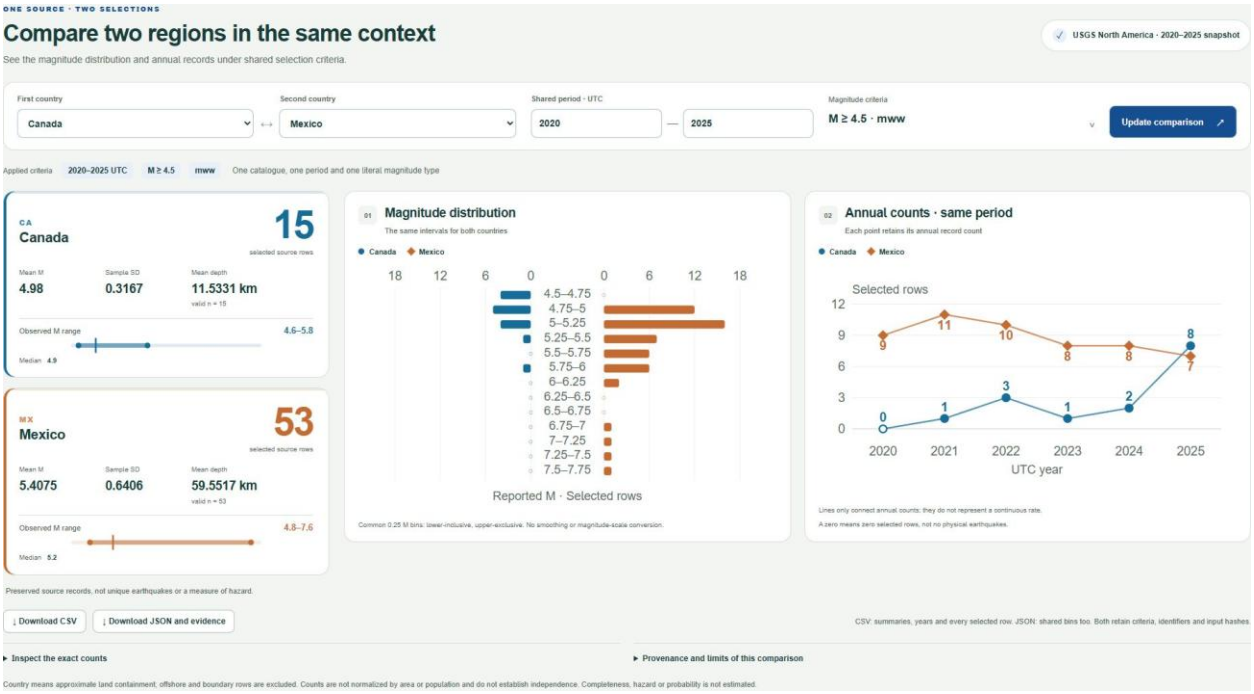


Figure 7. Canadian and Mexican land selections within USGS North America, UTC 2020-2025, M ≥ 4.5, literal mww: 15 and 53 rows. Common 0.25-M bins and annual counts preserve selection/evidence; they are not completeness-adjusted rates or relative hazard.

4.2. Guided historical consultation

Simple view exposes map, statistics and printable-report routes while retaining the complete workspace (Figure 8). The same regional predicate and export engine serve both interfaces; Help explains coverage and terms.

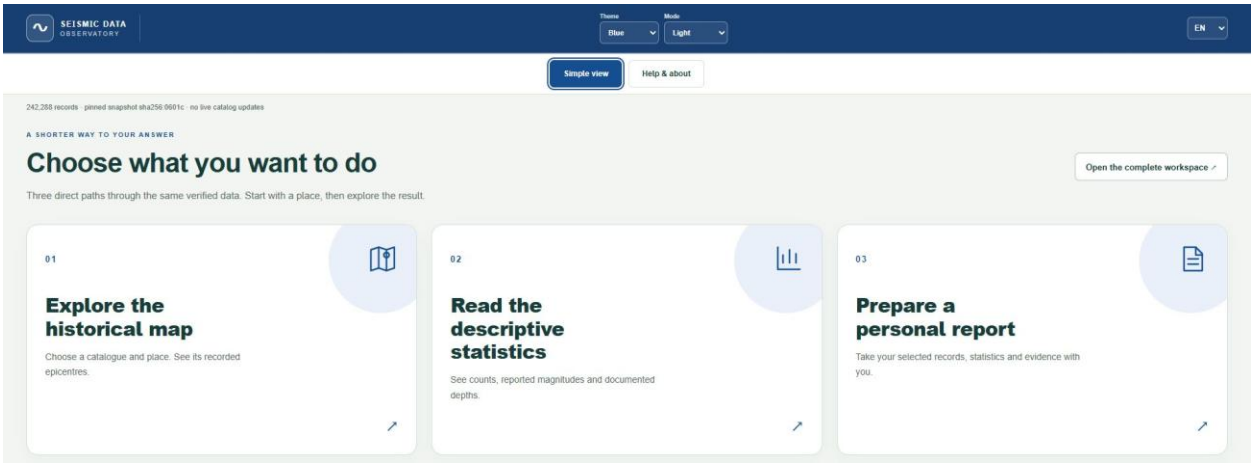


Figure 8. Three implemented Simple-view routes open historical maps, descriptive statistics and personalized reports through the complete workspace's filter/export engine. The full interface remains accessible. The capture demonstrates controls rather than measured usability gains.

The default Top 10 retains four-source measurements (Figure 9A). In advanced single-source mode, SSN's Guerrero land class contains 21,167 M ≥ 3.5 rows in UTC 1900-2026 (partial 2026, unknown type). Its Top 10 starts with M 7.8 on 28 July 1957. Fourteen rows satisfy M ≥ 7; a 50-km radius yields ten direct pairs involving twelve (Figure 9B), without establishing independent repeats or hazard. A separate Python geometric oracle checks real/synthetic ranking and distance cases.

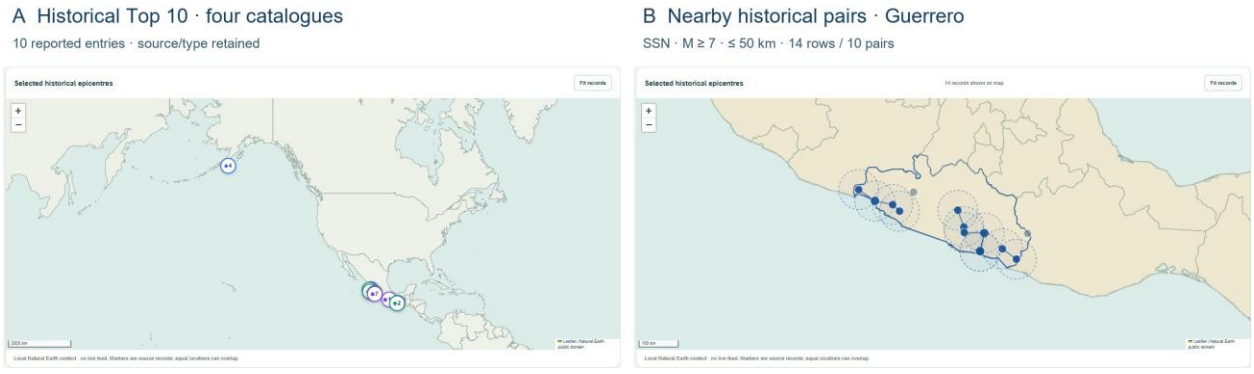


Figure 9. Complementary historical map extracts. (A) Default Top 10 ranks reported magnitude numbers across four catalogues, retaining source/type and both provenances for shared exact ComCat IDs; these are not homogeneous Mw measurements or reconciled physical events. (B) SSN Guerrero land, UTC 1900-2026 (partial 2026, unknown magnitude type): fourteen rows satisfy M ≥ 7; ten pairs at ≤50 km involve twelve. Offshore context is disabled. Links/circles are selection geometry, not rupture recurrence or risk. Tables and CSV/JSON preserve criteria and hashes.

4.3. Retained revisions and source roles

The pilot contains 22 events for 2017 and 14 for 2022. Its 236 public origin versions all have UPDATE status; primary US lineages retain 205 versions, not simultaneous active solutions (Table 3). Twenty-one ISC-GEM-authored origins explain the difference between US distribution and authorship; ten origins lack author attributes.

Table 3. Observed quantity	Value	Counting unit
Current-selected event identifiers	36	Captured events
Public origin-product versions	236	Retained versions
Versions in primary US origin lineages	205	Retained versions
Distributor source us	226	Public origin versions
Distributor source at / pt	6 / 4	Public origin versions
Origin author us / iscgem / missing	205 / 21 / 10	Public origin versions
Separate internal-origin / phase-data versions	15 / 203	Related products, excluded from primary analysis

Median endpoint displacement is 12.339 km, maximum 79.684 km; depth differences span -38.48 to 12.39 km. For us2000ahv0, retained endpoints change from 8.0 to 8.2 mww, 35.0 to 47.39 km depth and by 9.938 km epicentrally. These differences do not establish accuracy.

Thirteen of 31 same-type pairs change magnitude; five mixed-type pairs are excluded. Four cross 5.5 (Table 4), none the other tested thresholds. Among 30 mww pairs, earliest endpoints select 26 at $M \geq 5.5$ and latest select 30. Current-cohort admission at preferred $M \geq 5.5$ prevents interpreting upward crossings as an unbiased migration rate.

Table 4. Event identifier	Earliest retained mww	Latest retained mww
us2000ailq	5.4	5.5
us2000ap3b	5.4	5.6
us2000aqji	5.4	5.5
us1000arzf	5.4	5.5

The primary lineages contain 169 adjacent pairs: 117 change selected fields, 52 do not. Ten event-summary timestamps postdate every retained origin. In us2000ahv0, the primary lineage ends in December 2017, another branch updates in May 2022 and the October-2026 summary update relates to DYFI; these are different clocks, not one origin-revision sequence.

4.4. Validation and observed coverage

The 236 event-specific update boundaries yield 472 cutoffs and 4,720 queries, all agreeing with independent SQL. All 24 synthetic conditional projections also agree. Against known synthetic public counts, producer-time backcasting, filtering first and index-time substitution differ in 10, 11 and 10 cases. These overlapping constructed failures estimate neither operational error rates nor superiority. All stronger historical claims are declined by design, not through perfect detection of losses; empirical products contain no DELETE cases.

The full SSN download has records in 110 of 126 closed years (1900-2025), with 16 empty bins. Eight tied years produce 118 maximum-attaining rows, not 118 independent annual maxima. Partial 2026 contains 34,548 raw rows, 14,875 at $M \geq 3.5$ and 35 at $M \geq 5$. Coverage and reporting variations do not establish completeness, rates or return periods.

5. Discussion

The system turns heterogeneous catalogues into explicit study selections and reusable information, preserving source multiplicity rather than manufacturing a unique-event census. Exact magnitude types control one incompatibility but do not ensure constant measurement procedures. Land classes omit offshore events; optional nearby context preserves unassigned status rather than assigning jurisdiction. Spatial pairs, Pearson correlation and static plate context estimate neither rupture recurrence nor seismic risk.

Retained-version contracts provide weaker, reproducible calculations when historical public states are unknown. Existing history software supplies important prior capabilities; these contracts do not claim faster execution or independent accuracy gains. Synthetic counterexamples demonstrate logical ambiguity, while independent SQL validates calculations on supplied evidence.

Static files, local dependencies and standard outputs support migration to other hosts or local HTTP service. They do not guarantee twenty-year hosting or future browser compatibility. Memory limits and external tiles persist. Functional/responsive checks do not establish usability; independent users, broader cohorts and preserved historical

captures are needed. Probabilistic occurrence models remain future work requiring justified completeness, scale, sampling and stationarity assumptions.

6. Conclusions

The observatory connects browser computation, documented seismic inputs and traceable descriptive results. Coordinated maps, regional comparisons, historical selections and exports support subsequent study without a database server. Source roles, measurement types, geographic limits and retained-version contracts remain explicit. Tested calculations are reproducible; source accuracy, completeness, independent events and past public states remain unestablished. No earthquake forecast or validated seismic-risk estimate is produced.

Code and data availability

The evaluated interface is release 2026.11.10.2, distributed through the study repository (https://github.com/gabrielhuav/Seismic_Data_Geosciences) and its public explorer (https://gabrielhuav.github.io/Seismic_Data_Geosciences/). Software, manifests and frozen evidence accompany validation. Source hashes and capture receipts identify evaluated inputs and images; no archival DOI is asserted. The archive preserves 36 detail and four count/query responses; restore verified bytes rather than substitute live endpoints. SSN's full CSV SHA-256 is 0601c65544bc95ef865d3440e5e6d84249acdebd8edfb379669597096543f993. Its DOI identifies a mutable service, not this download. Credit belongs to SSN, Instituto de Geofísica, UNAM, without endorsement. Source-specific terms remain separate from MIT software, sql.js/SQLite notices, public-domain Natural Earth and ODC-BY PB2002 attribution.

Run python -m unittest discover -s tests and python -m scripts.validate_evidence; documented JavaScript/oracle checks validate browser calculations. Manifests distinguish observed inputs, geographic derivatives and synthetic controls.

CRediT authorship contribution statement

José Manuel Villa Vargas: Conceptualization; Formal analysis; Supervision; Writing - original draft; Writing - review & editing.

Gabriel Hurtado Avilés: Data curation; Investigation; Methodology; Project administration; Resources; Software; Validation; Visualization.

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Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI assistance

OpenAI Codex (OpenAI; GPT-6-based agent sessions, 6-10 November 2026) assisted literature discovery, code development, reproducible figure scripts, automated testing, manuscript drafting and document formatting. The explanatory diagram and data plots are rendered from documented inputs and maintained code. The interface panels derive from genuine integrated-browser captures of local previews, with capture-specific dates and hashes retained. Cropping and panel composition preserve the captured content; no generative image creation or retouching of observations is used.

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